Business Statistics

Relative location of observations



Some graphics and datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

Relative location of observations
Overview

- Describing the distribution of observations
- Chebyshev's Theorem
- The Empirical Rule
- z-Score

Relative location of observations

As functions of the standard deviation

Relative location of observations
How are the values of a variable distributed?

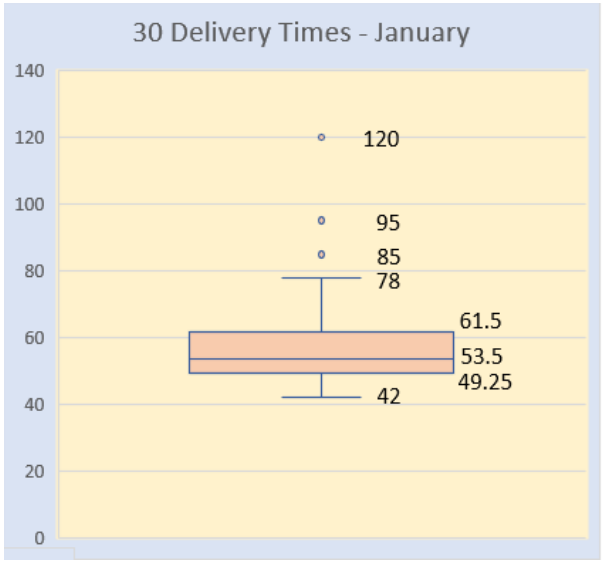
- There are measures beyond the standard deviation to help understand how the values of a variable are distributed
- Distributions and locations can be in terms of the standard deviation.
  - This might be “within 2 standard deviations”
  - Or “located 1.3 standard deviations above the mean.”
- With the standard deviation, we can find out:
  - What is the minimum proportion of observations within k standard deviations from the mean?
  - What is the relative position of a single observation in terms of standard deviations

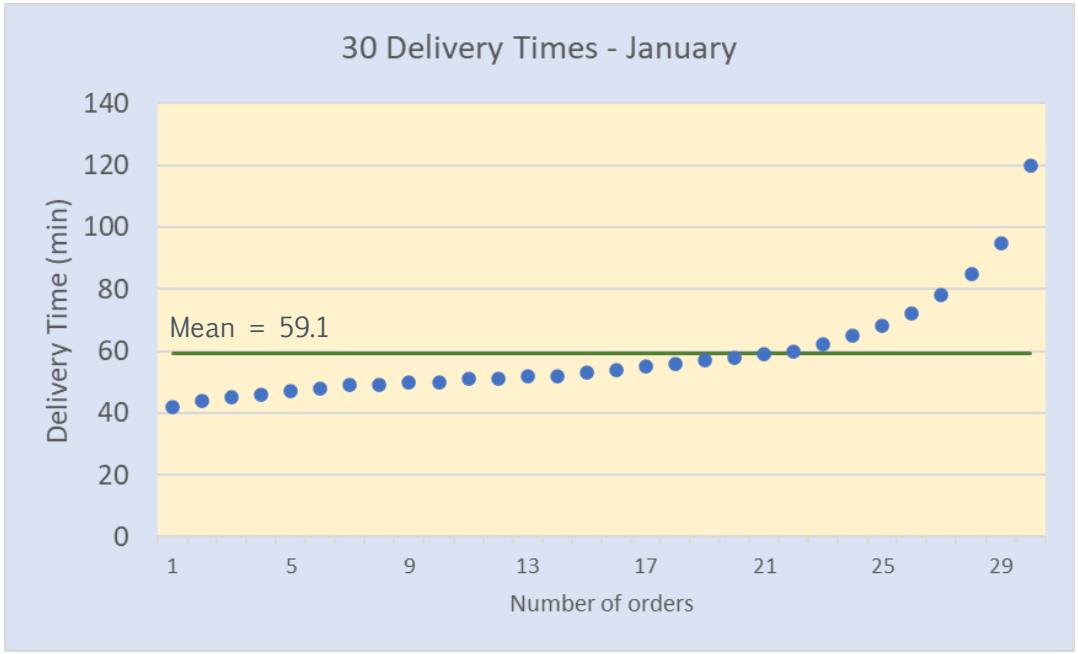
Relative location of an observation
Data set and ordered measures

- An online retailer tracks the delivery time (in minutes) for 30 recent customer orders shipped from the same warehouse in January.
- Management wants to understand typical delivery time, variability, consistency, and whether extreme delays exist.

Delivery Time
42
48
51
59
54
…
50
53
58
68
120

5 Number Summary January				
Min	Q1	Q2	Q3	Max
42	49.25	53.5	61.5	120





Relative location of observations

Chebyshev's Theorem

Relative location of observations
Chebyshev's Theorem for all variables

- More information about the location of observations is possible once the standard deviation is available

- For **any** variable, **Chebyshev's Theorem** tells us that the minimum proportion of observations included within k standard deviations of the mean is $\left(1-\frac{1}{k^2}\right)\%$, for values of $k > 1$
  - For $k = 1\ SD$, $\left(1-\frac{1}{k^2}\right) = 0\%$ of observations, which is not right

- At least $\left(1-\frac{1}{k^2}\right)\%$ of observations are within k standard deviations of the mean

k Std. Dev.	$\left(1-\frac{1}{k^2}\right)\%$ of observations
2	$\left(1-\frac{1}{2^2}\right) = \frac{3}{4} = 75\%$
3	$\left(1-\frac{1}{3^2}\right) = \frac{8}{9} = 89\%$
4	$\left(1-\frac{1}{4^2}\right) = \frac{15}{16} = 94\%$

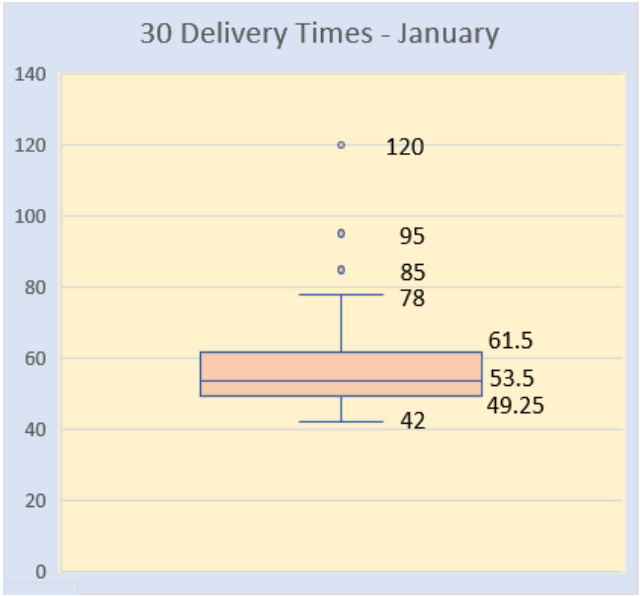
Relative location of observations
Chebyshev's Theorem: delivery time in minutes (mins)

- At least $\left(1-\frac{1}{k^2}\right)\%$ of obs. are within k standard deviations of the mean

- For the delivery times data:
  - n = 30
  - $\overline{x}$ = 59.1
  - SD = 16.8
  - (min, max) = (42, 120)

5 Number Summary January				
Min	Q1	Q2	Q3	Max
42	49.25	53.5	61.5	120

k Standard Deviations	Mean - Std. Dev.	Mean + Std. Dev.	% Minimum Observations	Minimum Observations
1	42.3	75.9	-	-
2	25.5	92.7	75%	23
3	8.7	109.5	89%	27
4	-8.0	126.2	94%	28



Relative location of an observation Nonparametric methods

- Chebyshev's Theorem applies to all variable distributions
- That means there are no assumptions or restrictions placed on a variable's distribution for the theorem to be valid
- Therefore, this is called a "**nonparametric**" or "distribution free" method
- A distribution free method must allow for all different distributions and therefore cannot take advantage of characteristics of specific types of distributions
- Methods made for specific types of variable distributions can take advantage of the specific shape and spread to give more specific information

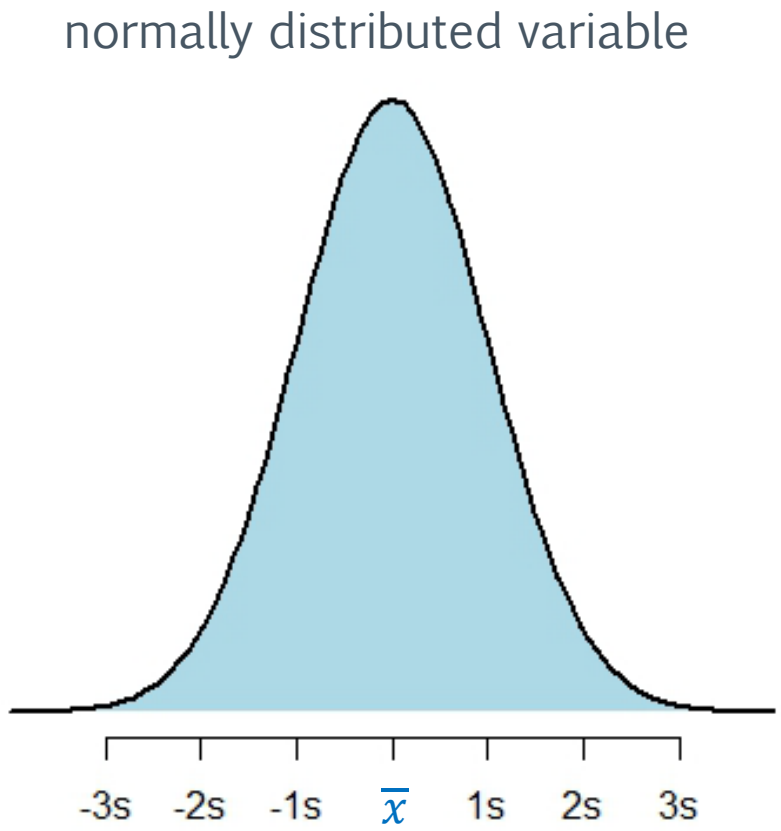
Relative location of observations

The Empirical Rule

Relative location of observations
The Empirical Rule for normal distributions

- For normally distributed variables, the **Empirical Rule** provides the number of observations located within +/- 1, 2, and 3 SDs of the mean

- This Empirical Rule provides a good **approximation** for **any distribution** that is roughly symmetric and bell-shaped



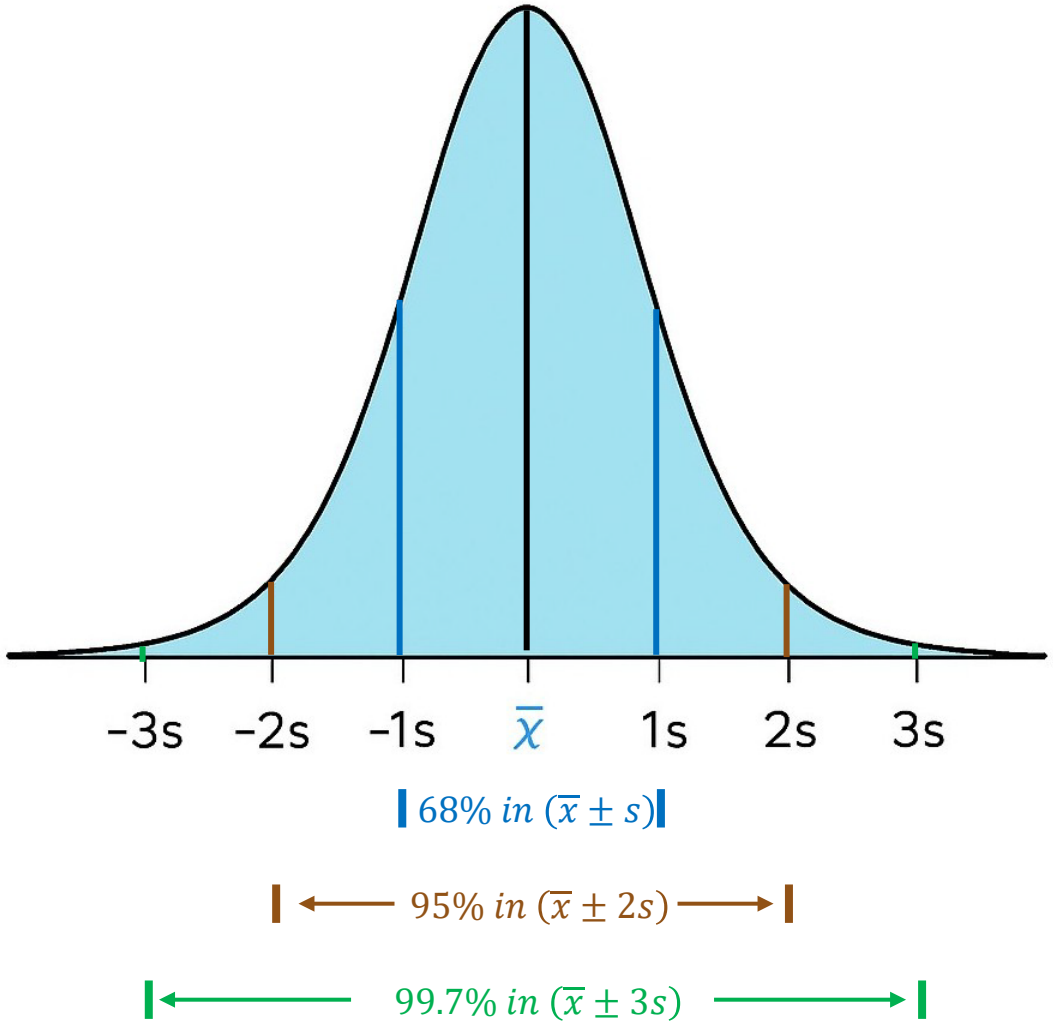
Relative location of observations
The Empirical Rule: 68 95 99 rule

- The Empirical Rule provides these approximations:
  - The interval $(\bar{x} \pm s)$ contains approximately 68% of all the observations
  - The interval $(\bar{x} \pm 2s)$ contains approximately 95% of all the observations
  - The interval $(\bar{x} \pm 3s)$ contains approximately 99.7% of all the observations
  - The corresponding population formulas are $(\mu \pm \sigma), (\mu \pm 2\sigma), (\mu \pm 3\sigma)$

- Note that Chebyshev's Theorem had no estimate for the proportion of observations within +/- 1 SD of the mean

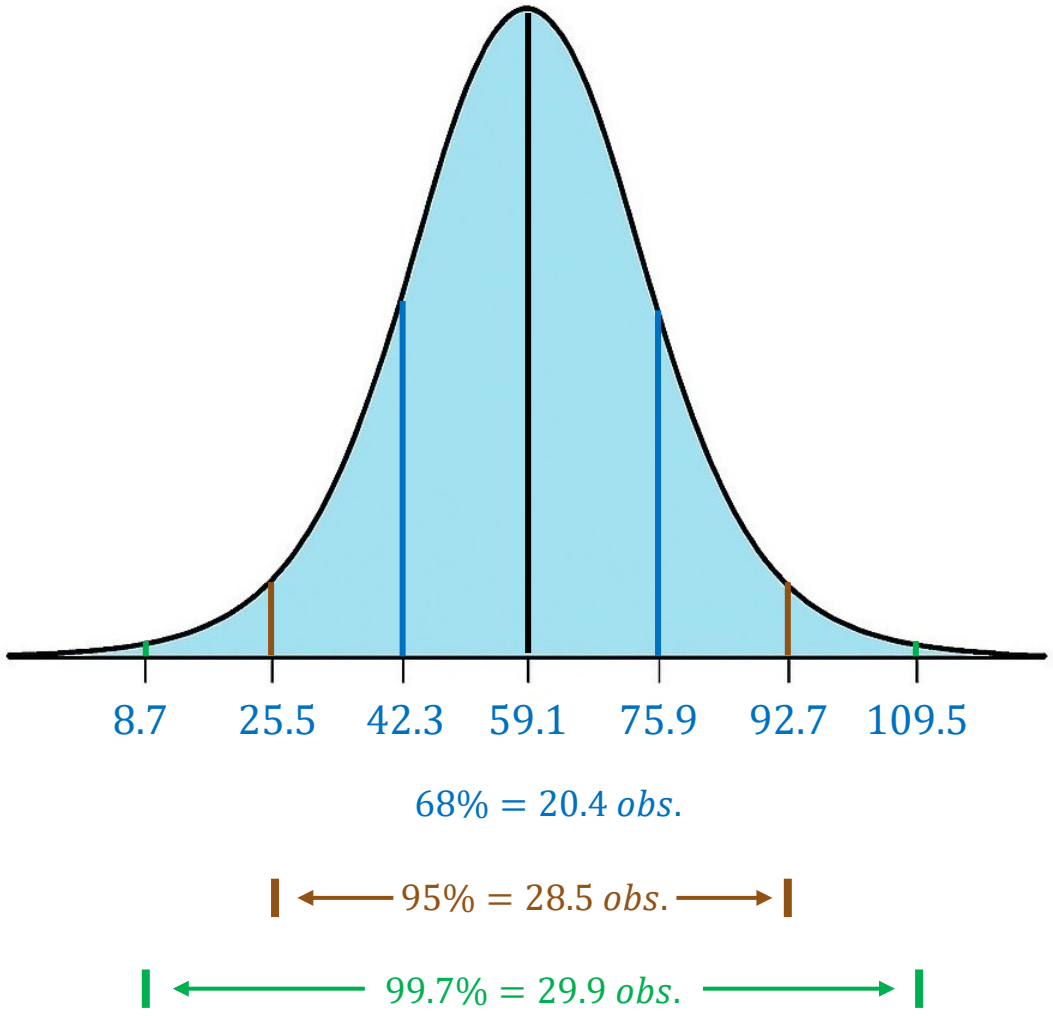
Sample Interval	Population Interval	% of observations contained
$(\bar{x} \pm s)$	$(\mu \pm \sigma)$	68%
$(\bar{x} \pm 2s)$	$(\mu \pm 2\sigma)$	95%
$(\bar{x} \pm 3s)$	$(\mu \pm 3\sigma)$	99.7%

Relative location of observations
The Empirical Rule: 68 95 99 rule



Sample Interval	% of observations contained
$(\bar{x} \pm s)$	68%
$(\bar{x} \pm 2s)$	95%
$(\bar{x} \pm 3s)$	99.7%

Relative location of observations
The Empirical Rule: delivery time in minutes (mins)



n = 30

$\overline{x}$ = 59.1

s = 16.8

Interval	Interval	% of obs.	obs. contained
(59.1 ± 16.8)	$(42.3, 75.9)$	68%	20.4
(59.1 ± 33.6)	$(25.5, 92.7)$	95%	28.5
(59.1 ± 50.4)	$(8.7, 109.5)$	99.7%	29.9

Relative location of observations

z-score

Relative location of an observation
z-score

- The empirical rule and Chebyshev's Theorem provide information about **the distribution** - how many observations were in certain intervals
- The z-score provides the relative location of an **individual observation**
- For a given set of data, the z–score transforms an observation value into the number of standard deviations that value is from the mean

z-score for a sample

x = observation value
$\bar{x}$ = sample mean
s = sample standard deviation

$$z = \frac{x - \bar{x}}{s}$$

z-score for the population

x = observation value
μ = population mean
σ = population standard deviation

$$z = \frac{x - \mu}{\sigma}$$

Relative location of observations
z-score: delivery time in minutes (mins)

$\bar{x}$ = 59.1

s = 16.8

$$z = \frac{x_i - \bar{x}}{s}$$

x	$z = \frac{x - 59.1}{16.8}$
42	-1.02
59	-0.01
95	2.14

